

Kaan Aytekin

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EDUCATION

University of Rochester

Bachelor of Science, Major in Mechanical Engineering

- GPA 3.94 out of 4.00, Dean's List all applicable semesters
- Davis UWC Scholar, Tau Beta Pi Scholar

Rochester, NY

Anticipated May 2026

UWC Robert Bosch College

IB Diploma

Freiburg, Germany

May 2022

WORK EXPERIENCE

Laboratory for Laser Energetics

Research Assistant

Rochester, NY

March 2023 - Present

- Conducted in-depth numerical calculations of free-free laser absorption rates in plasmas, incorporating quantum scattering theory and screening effects.
- Leveraged Python programming skills (NumPy, SciPy and Matplotlib), to perform numerical evaluations of Ordinary Differential Equations (ODEs) and integrals.
- Performed computations of 100 absorption correction tables (Gaunt Factor) using a Linux-based HPC cluster. Fitted a parametric analytical expression with a three-variable equation to extract data for integration into opacity and hydro-codes.
- Presented a poster in the student session of APS DPP 2024 and received the Outstanding Poster Award.

RESEARCH PROJECTS

Determining Yield Characteristics of Heat Treated 4130 Steel Under Torsion

Rochester, NY

Author

January 2024 - May 2024

- Developed a custom torque testing rig and testing methodology to measure the torsional yield characteristics of heat-treated 4130 steel (no-treatment, annealed and quenched samples) using strain gauges, analyzing strain data using MATLAB and Mohr's Circle.
- Conclusively validated that targeted heat treatment statistically and significantly improves the torsional yield point of steel half-shaft connections, supporting component weight reduction in vehicle design.

Aerodynamic CFD Based Optimization for Taxi Cars using Bezier Curves

Freiburg, Germany

Author

February 2021 - October 2021

- Achieved an 11% reduction in taxi car drag coefficient by leveraging adjoint optimization methodologies within ANSYS Fluent, following initial computational fluid dynamics (CFD) simulation and Bezier curve-based geometric modeling in Fusion 360.

ACTIVITIES

UR Baja SAE (Offroad Car Collegiate Design Series)

Rochester, NY

Chief Engineer and Drivetrain Project Team Co-Leader

May 2024 - May 2025

- Directed a 30-member Baja SAE team through the complete vehicle design and fabrication, achieving a 40% increase in competition rankings while reducing the project budget by 5%.
- Designed the team's first-ever custom differential along with a custom transfer case and driveshaft, leveraging Siemens NX for all component modeling and assembly.
- Utilized AGMA equations and Finite Element Analysis (FEA) to compute the structural integrity and loads from bevel gears, applying those loads for subsequent analysis of the custom shafts and case.

Power Distribution Project Team Leader (Responsible of Axles)

August 2023 - May 2024

- Optimized axle and universal joint geometry using FEA (NASTRAN), dynamics simulations (Siemens Simcenter 3D), and solid mechanics principles, resolving critical interference and loosening issues while achieving a 64% reduction in material costs.
- Conducted non-linear finite element analysis (NASTRAN Sol. 402) on axle components and validated the results through a custom torque testing rig to ensure simulations represent reality.
- Manufactured 50+ custom designed axle yokes in a 4-axis CNC mill, used a manual mill to manufacture axle flanges, inserts and more.

AWARDS

APS DPP '24 Outstanding Poster Award

Atlanta, GA

American Physical Society Division of Plasma Physics Annual Meeting

October 2024

- Awarded to the top 5 student presenters out of approximately 150 presenters.

SKILLS & INTERESTS

- Siemens NX, Python (NumPy, SciPy, Matplotlib), MATLAB, NASTRAN, Fusion 360, SolidWorks, PTC Creo, Ansys Fluent (Workbench, Space Claim, Mesh), L^AT_EX, Office Suite, Adobe Photoshop, Adobe Lightroom, DaVinci Resolve
- Turkish (Native), English (Fluent), German (B1)
- Music, played and recorded piano and guitar, led band as the club leader and the lead guitarist.